Reconstructing Intra-Tumor Fitness Landscapes from scSeq CNA Genotypes via Simulation-Based Bayesian Inference and Deep Learning

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BACKGROUND

Clone fitness has so far been measured by **watching** a tumour evolve: repeated single-cell sequencing of cell lines and xenografts, then time-series modelling (fitClone, Salehi et al. 2021). A patient, by contrast, is usually sequenced once: a **single snapshot** of the tumour, not a time series.

Question

From one clonal copy-number snapshot, can we recover the selection coefficient of every chromosome arm, $\theta \in \mathbb{R}^{44}$, with its uncertainty? There is **no likelihood** — tumour evolution exists only as a simulator — and **extinct clones are unseen**.

Why arm-level θ?

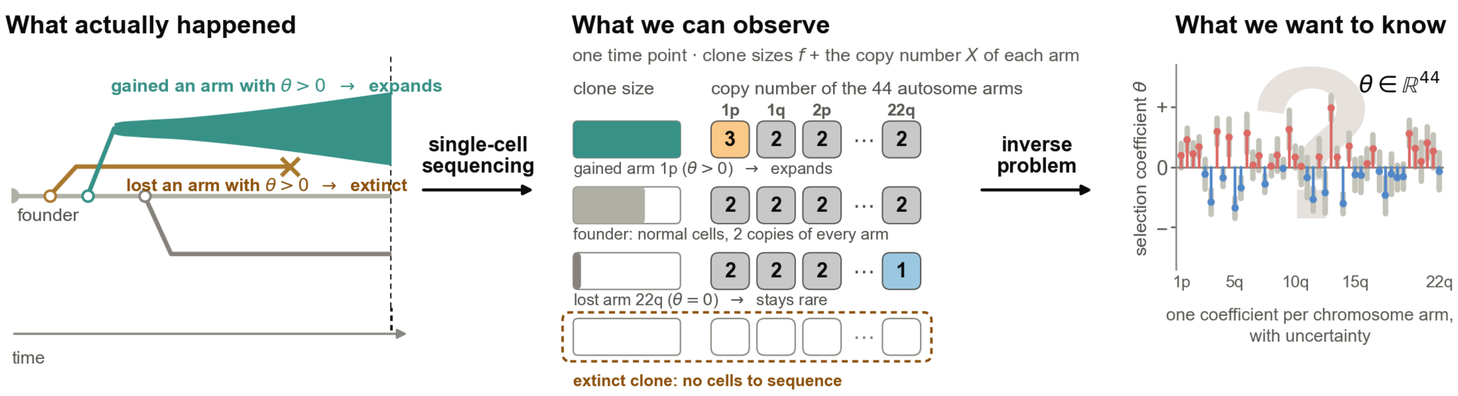
A cell's **fitness** — its odds of dividing rather than dying — is set by its copy numbers:

$$\text{fitness} = \prod_{\text{arms}} (1 + \theta_{\text{arm}})^{\text{copies / ploidy}}$$

θ_{arm} is the weight of one extra copy of that arm: $\theta > 0$ favours gains, $\theta < 0$ favours losses. Fitness is a property of *one clone*; θ is a property of the *arm*, so it

- **scores any clone** from its copy numbers — including clones not yet seen;
- **predicts the future** of the tumour: which clone takes over, what a therapy leaves behind;
- **ranks the arms** that drive this cancer — the fitness landscape, comparable across patients;
- **comes with uncertainty**: the posterior says which arms are established and which are guesses.

What actually happened, what one biopsy shows, and what we want to know

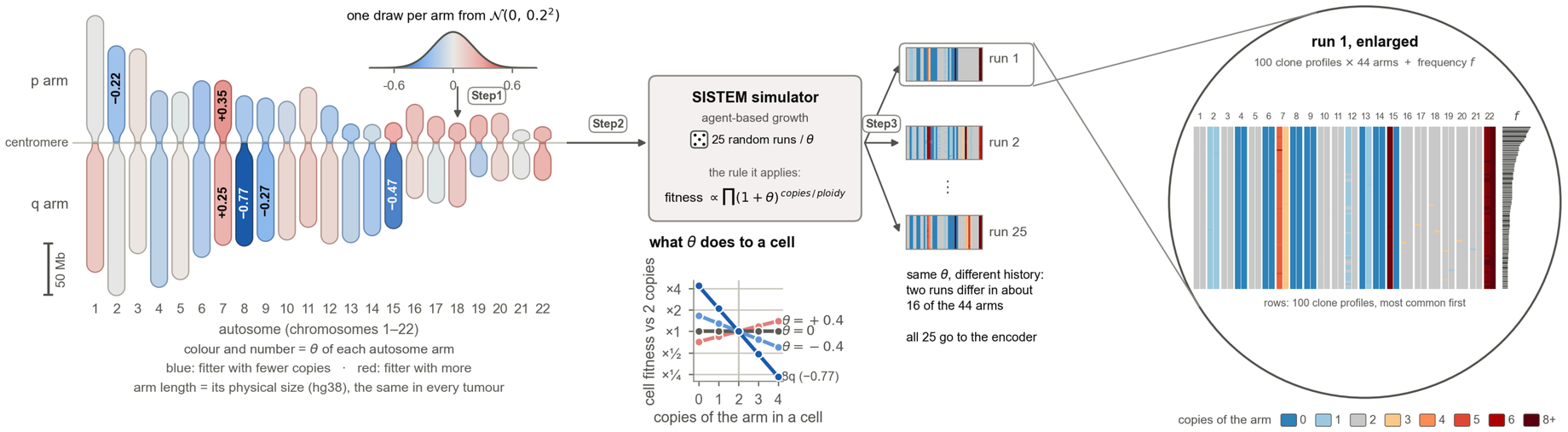


THIS WORK

1. simulate thousands of tumours with **known θ**
2. train a **neural posterior** on them
3. read **θ and its uncertainty** from a single snapshot

DATA: SIMULATED TUMOURS

Step1: draw one θ per autosome arm → **Step2:** SISTEM grows tumours under it → **Step3:** 25 replicate tumours; the 100 most common clone profiles of each tumour and their frequencies f are what the model sees

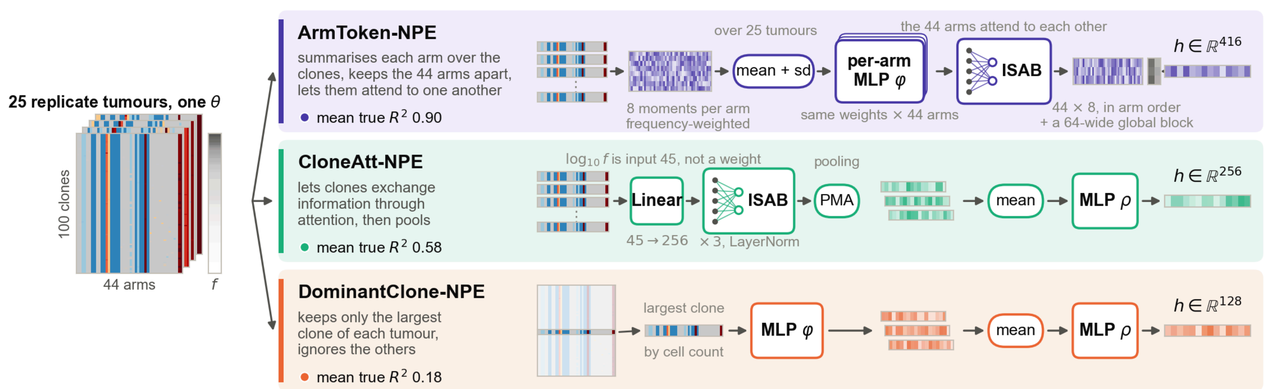


DATA

- **Simulated:** 2,500 simulator parameter settings × 25 independent replicates = 62,500 tumours; split 80 % / 20 % by parameter setting into training and test sets.
- **Parameters:**
  - 44 chromosome-arm selection coefficients $\theta \sim \mathcal{N}(0, 0.2)$ — the inference targets
  - two nuisance CNA rates: $\text{arm_rate} \sim \text{Unif}(10^{-5}, 10^{-4})$, $\text{chromosomal_rate} \sim \text{Unif}(10^{-6}, 10^{-5})$
  - fixed: 5 Mb regions, focal CNA rate 5×10^{-4}, growth to 5×10^{6} cells
- **Selection:** at the chromosome-arm level, as in SISTEM's arm model: one θ per arm, no fitness for genes or point mutations; focal CNAs act only via the arm's average copy number.
- **Representation:** per clone, the arm-level copy-number ratio of the 44 autosomal arms, normalized as $\log_2(\max(x, 10^{-3})) - 1$; cells sharing a profile form a clone, and the 100 most frequent clones are kept, each with its relative frequency → a 100 × 45 matrix X.

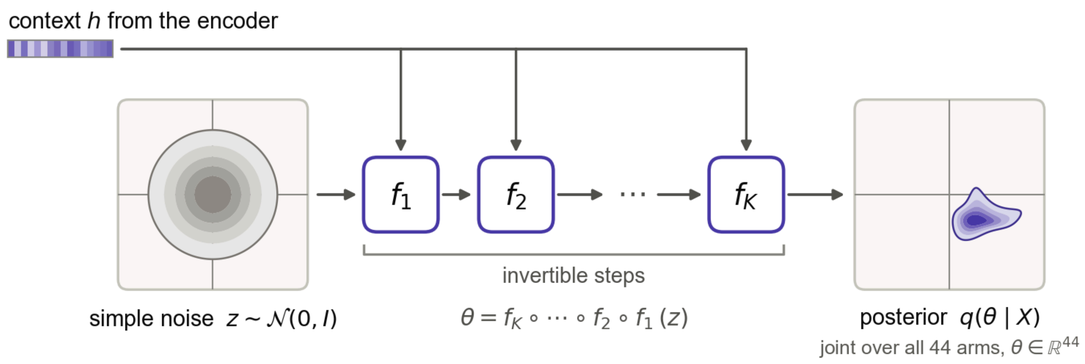
METHODS: ENCODERS

three ways to compress the 25 tumours into one context vector h: **CloneMLP** (every clone, frequency-weighted mean), **CloneAtt** (Set Transformer), **DominantClone** (largest clone only)



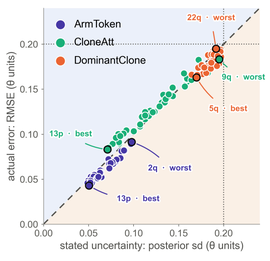
METHODS: POSTERIOR

a neural spline flow (K = 5 invertible steps, sbi) turns noise into q(θ | X), steered by h · trained on simulations only · amortised: instant for a new tumour



RESULTS: UNCERTAINTY VS ERROR

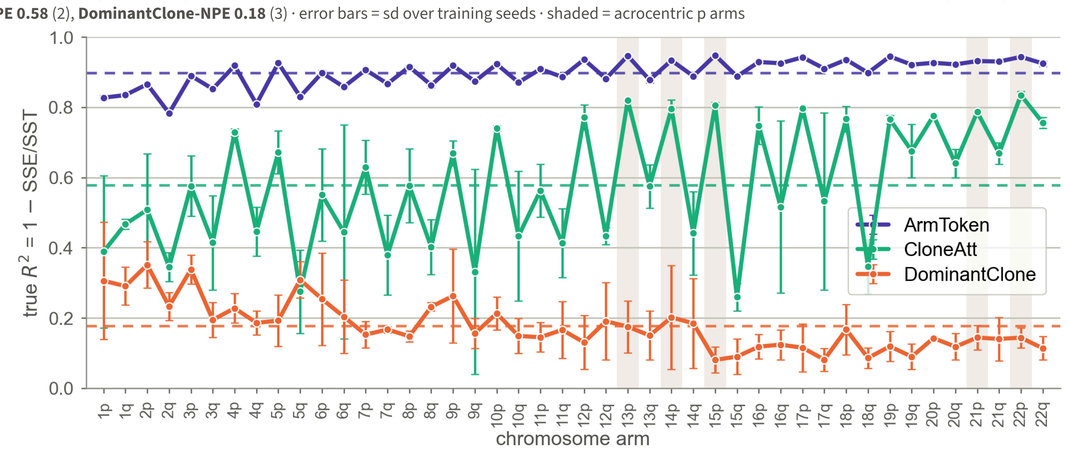
one dot per arm, both axes in θ units · dashed = honest (stated sd equals real error) · dotted = prior sd 0.20



- **ArmToken-NPE** — stated sd 0.067 · error 0.061 · ratio 0.90 · best 13p, worst 2q
- **CloneAtt-NPE** — 0.123 · 0.128 · ratio 1.04 · best 13p, worst 9q
- **DominantClone-NPE** — 0.187 · 0.182 · ratio 0.97 · best 5q, worst 22q: the prior handed back
- ratio 1 = the model's uncertainty matches its error; above the line = over-confident (error bars too narrow), below = under-confident (too wide); nearer the origin = more learned

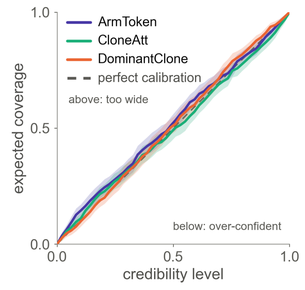
RESULTS: RECOVERY PER ARM

per-arm true R^2 (1 − SSE/SST) on 651 held-out simulated tumours · dashed = each encoder's mean over the 44 arms: **ArmToken-NPE 0.90** (3 seeds), **CloneAtt-NPE 0.58** (2), **DominantClone-NPE 0.18** (3) · error bars = sd over training seeds · shaded = acrocentric p arms



RESULTS: CALIBRATION

joint coverage over all 44 arms (TARP) · diagonal = honest



- **ArmToken-NPE** — SBC 21/44 arms fail (15–37 over seeds) · 95 % cov. 0.961 · TARP 0.022, slightly wide
- **CloneAtt-NPE** — 28/44 (26–28) · 0.930 · TARP 0.016, slightly over-confident
- **DominantClone-NPE** — 9/44 (8–10) · 0.954 · TARP 0.011, slightly wide

RESULTS

- **The clone set carries the signal:** the largest clone alone recovers θ with true R^2 0.18; attention over all clones 0.58.
- **Sharing weights across arms unlocks it:** ArmToken-NPE reaches 0.90 (± 0.002 over 3 seeds), ahead on 44 of 44 arms; the published model scored 0.47 with over-confident intervals.
- **Honest uncertainty:** all three encoders sit on the TARP diagonal; ArmToken's 95 % intervals cover 96 % and are 4× narrower than the prior.
- **Arms differ:** p arms beat q arms, and accuracy rises with chromosome number — but falls for the largest-clone baseline.
- 651 held-out simulated tumours; seed noise ±0.002 (ArmToken), ±0.015 (CloneAtt), ±0.017 (DominantClone).

MORE INFORMATION

Preprint bioRxiv 10.64898/2026.04.10.717795

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Acknowledgments This study was partially supported by NSF grants 2415564 and 2415562.

DISCUSSION

- **Feasible:** arm-level selection can be read from a single clonal snapshot, with uncertainty, without a likelihood.
- **Caveats:** the model was trained and tested with 25 replicate tumours per θ — a patient gives one, so it must be evaluated with a single tumour; the inputs are exact simulator copy numbers, whereas scDNA-seq samples cells and calls copy numbers with noise — simulating that step is the next thing to add.
- **Next:** test the model on real tumours.

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